

SPHERE: MISSION CONTROL

STUDENT LABORATORY TASK SHEET & EXPERIMENTAL REGISTER

STUDENT NAME(S):

TEAM / TEST IDENTIFIER:

DATE / LABORATORY PERIOD:

ASSIGNED TPS CONFIGURATION:

[ ] A: Bare [ ] B: Bubble wrap [ ] C: Mylar [ ] D: MLI Composite

PART A. EXPERIMENTAL TEMPERATURE LOG (0 TO 15 MIN)

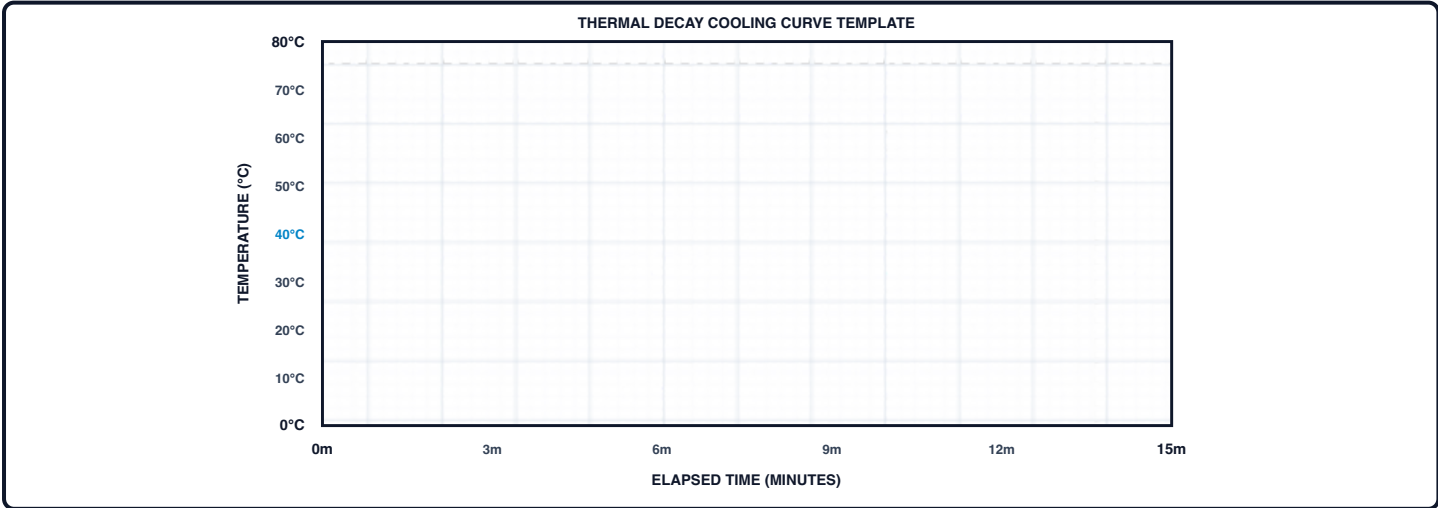
60s INTERVALS

| ELAPSED TIME | OBSERVED TEMP ( $T_{obs}$ ) | MODEL TEMP ( $T_{model}$ ) | RESIDUAL $ \Delta T $ (°C) | OBSERVER FIELD NOTES & BOUNDARY STATUS            |
|--------------|-----------------------------|----------------------------|----------------------------|---------------------------------------------------|
| 0 min        | 80.0°C                      | 80.0°C                     | 0.0°C                      | Capsule submerged; ice bath boundary verified.    |
| 1 min        |                             |                            |                            | Check immersion depth and probe position.         |
| 2 min        |                             |                            |                            |                                                   |
| 3 min        |                             |                            |                            |                                                   |
| 4 min        |                             |                            |                            |                                                   |
| 5 min        |                             |                            |                            | Mid-phase checkpoint; verify ice slurry volume.   |
| 6 min        |                             |                            |                            |                                                   |
| 7 min        |                             |                            |                            |                                                   |
| 8 min        |                             |                            |                            | Record any change in bath condition.              |
| 9 min        |                             |                            |                            |                                                   |
| 10 min       |                             |                            |                            | Check for leakage or probe contact with the wall. |
| 11 min       |                             |                            |                            |                                                   |
| 12 min       |                             |                            |                            |                                                   |
| 13 min       |                             |                            |                            |                                                   |
| 14 min       |                             |                            |                            |                                                   |
| 15 min       |                             |                            |                            | Final reading; record bath temperature.           |

PART B. THERMAL COOLING CURVE GRAPH GRID

EMPIRICAL (•) VS THEORETICAL (---)

Plot measured temperatures as solid points and connect them with a line. Plot the analytical reference-model values as a dashed line.



PART C. QUANTITATIVE RESIDUAL & ERROR ANALYSIS

STATISTICAL EVALUATION

1. Sum of Absolute Residuals:

$\sum |\Delta T| = \sum |T_{\text{obs}} - T_{\text{model}}|$

$\sum |\Delta T| = \text{-----}^{\circ}\text{C}$

2. Mean Absolute Error (MAE):

$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |T_{\text{obs},i} - T_{\text{model},i}|$

$\text{MAE} = \text{-----}^{\circ}\text{C}$

3. Interpretation:

Compare MAE with the thermometer accuracy and variation between repeated measurements.

Number of observations:  
-----

MODEL-AGREEMENT RESULT:

MAE = ----- $^{\circ}\text{C}$

☐ uncertainty discussed

☐ model assumptions discussed

PART D. POST-FLIGHT SCIENTIFIC INQUIRIES & ANALYSIS

6 CORE CONCEPTS

Inquiry 1: Radiant Barriers vs. Direct Conduction

Reflective film can reduce radiative exchange when it faces a suitable gap. Why may it provide limited insulation when placed directly in ice water without a low-conductivity air or fibre spacer?

Inquiry 2: Understanding the Decay Constant (*k*)

In Newton's Law  $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$ , does a smaller *k* value mean better or worse insulation? Explain how *k* controls the cooling speed.

Inquiry 3: Why Multi-Layer Insulation (MLI) Works

Why can the multilayer configuration reduce the cooling rate relative to a single layer? Discuss trapped air, fluid motion, radiation and interfacial resistance.

Inquiry 4: Real-World Experimental Factors

What physical factors in the lab might cause your sensor readings to drift away from the computer simulation? Name at least two specific factors (e.g. sensor depth, stirring, seal leaks).

Inquiry 5: Ice Bath Temperature Drift

If the ice bath warms up from 0.0 $^{\circ}\text{C}$  to 4.5 $^{\circ}\text{C}$  during your 15-minute run, how should you update the environment temperature  $T_{\text{env}}$  in Mission Control so the model stays accurate?

Inquiry 6: Effect of Capsule Water Volume

If you doubled the water volume from 100 mL to 200 mL, how would that extra thermal mass ( $Q = mc_p\Delta T$ ) change the cooling rate and the value of *k*?

STUDENT SIGNATURE:

\_\_\_\_\_

INSTRUCTOR VALIDATION:

\_\_\_\_\_

EVALUATION SCORE: \_\_\_\_\_ / 100